Compton Effect: particle nature of light

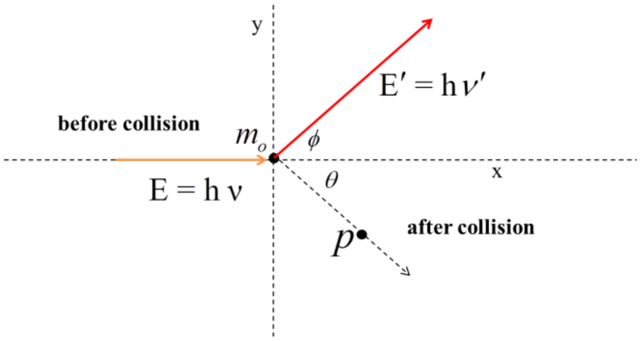

Fig.1

Let us consider a photon of energy $E = h\nu$ incident on an electron having rest mass m_0 and which is initially at rest. After the collision the photon having energy $E' = h\nu'$is scattered at an angle ϕ and an electron moves with momentum p and making and angle θ from the x axis.

Now according to energy conservation principle

$$Loss\ in\ photon\ energy = gain\ in\ kinetic\ energy\ of\ the\ electron$$

$$h\nu - h\nu' = KE \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$$

Applying momentum conservation along x and y directions

$$\frac{h\nu}{c} = \frac{h\nu'}{c}\cos\phi + p\cos\theta$$

$$pc\cos\theta = h\nu - h\nu'\cos\phi \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

$$pc\sin\theta = h\nu'\sin\phi \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots.. (3)$$

Now squiring and adding equations (2) and (3)

$$p^2c^2 = (h\nu)^2 + (h\nu')^2 - 2h^2\nu\nu'\cos\phi \ldots\ldots\ldots\ldots\ldots (4)$$

Now since the total energy of the particle is defined as

$$E = KE + m_0c^2 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(5)$$

Relation between total energy, momentum and rest energy of the particle

$$E^2 = p^2c^2 + m_0^2c^4 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(6)$$

Substituting the value of equation (1) in equation (5)

$$E = (h\nu - h\nu') + m_0c^2 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots..(7)$$

Now take the square of equation (7) and substituting in (6), we have

$$\{(h\nu - h\nu') + m_0c^2\}^2 = p^2c^2 + m_0^2c^4$$

$$(h\nu)^2 + (h\nu')^2 - 2h^2\nu\nu' + {m_0}^2c^4 + 2hm_0c^2(\nu - \nu') = p^2c^2 + m_0^2c^4$$

$$(h\nu)^2 + (h\nu')^2 - 2h^2\nu\nu' + 2hm_0c^2(\nu - \nu') = p^2c^2 \ldots\ldots\ldots\ldots(8)$$

Now comparing equations (4) and (8) we have

$$(h\nu)^2 + (h\nu')^2 - 2h^2\nu\nu' + 2hm_0c^2(\nu - \nu') = (h\nu)^2 + (h\nu')^2 - 2h^2\nu\nu' cos\phi$$

$$2hm_0c^2(\nu - \nu') = 2h^2\nu\nu' - 2h^2\nu\nu' cos\phi$$

$$m_0c^2(\nu - \nu') = h\nu\nu'(1 - cos\phi)$$

$$\frac{(\nu-\nu')}{\nu\nu'} = \frac{h}{m_0c^2}(1 - cos\phi)\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(9)$$

Since $c = \nu\lambda$

$$\nu = \frac{c}{\lambda}$$

$$\frac{1}{\nu} = \frac{\lambda}{c}$$

$$\Delta\lambda = (\lambda' - \lambda) = \frac{h}{m_0c}(1 - cos\phi)$$

Where $\boldsymbol{\lambda_C = \frac{h}{m_0c}}$ **is known as Compton wavelength.**

$$\boldsymbol{\lambda_C = \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-31} \times 3.0 \times 10^{8}} = 2.426 \times 10^{-12} meter = 2.426\, pm}$$